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Taming Noise-Induced Prototype Degradation for Privacy-Preserving Personalized Federated Fine-Tuning

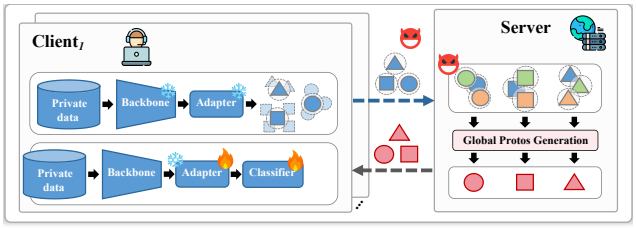
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Abstract

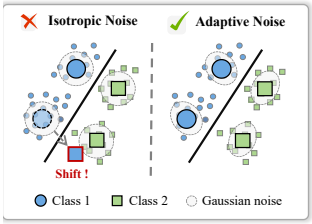
Prototype-based Personalized Federated Learning (ProtoPFL) enables efficient multi-domain adaptation by communicating compact class prototypes, but directly sharing them poses privacy risks. A common defense involves per-example ℓ_2 clipping before prototype computation to bound sensitivity, followed by isotropic Gaussian noise to enforce Local Differential Privacy (LDP). However, Isotropic Gaussian Prototype Perturbation (IGPP) typically over-perturbs discriminative dimensions and struggles to balance the clipping threshold with representation fidelity. In this paper, we propose ***VPDR****, a client-side privacy plug-in that seamlessly integrates into existing ProtoPFLs. Motivated by the observation that dimension-wise class variance reflects discriminability, we introduce* ***Variance-adaptive Prototype Perturbation (VPP)****, which allocates less noise to discriminative subspaces, preserving semantic separability while ensuring privacy. We further develop* ***Distillation-guided Clipping Regularization (DCR)****, which enables feature norms to adaptively concentrate near the predefined clipping threshold while maintaining prediction consistency. Theoretical analysis shows that our groupwise mechanism provides privacy guarantees no weaker than the isotropic baseline under the same privacy constraints. Extensive experiments on multi-domain benchmarks demonstrate that VPDR achieves a superior privacy-utility trade-off, outperforming IGPP in personalized federated fine-tuning without sacrificing robustness against realistic attacks.*

1. Introduction

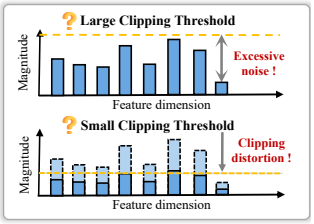
Scaling laws for foundation models have been established on massive public datasets [1]. As general pre-training has largely consumed available public data [39], leveraging private data becomes essential to overcome the emerging data-scarcity bottleneck [25]. Federated Learning (FL) [2, 19, 23, 31] offers a distributed paradigm for collaboratively fine-tuning pre-trained models without sharing raw data [50, 56]. Under domain skew, however, a single global model often underperforms or even fails to converge across clients [4, 40, 45]. To solve this, Personalized FL (PFL) [10, 21, 34, 47] adapts shared models with client-specific components [22, 34, 35, 49, 51]. In PFL, prototype-based methods [11, 15, 36, 42, 52, 53] exchange compact class statistics such as class means or centroids to align global and local semantics, enabling lightweight personalization while maintaining frozen backbones.



(a) ProtoPFL with IGPP.



(b) Noise vs. Discriminability.



(c) Clipping-Threshold Dilemma.

Figure 1. **Motivation illustration**. In (b), the dashed circle shows uniform noise and the dashed ellipse our adaptive noise. Isotropic noise shifts the blue Class 1 prototype into the red Class 2 region. In (c), a generous clipping bound leaves feature norms almost unchanged but forces a large noise scale, while an aggressive bound shrinks many features, causing severe information loss.

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Despite their effectiveness, directly sharing prototypes poses significant privacy risks [43]. Since prototypes condense features into domain fingerprints, they can closely resemble individual instances under small sample sizes, opening the door to membership inference and reconstruction attacks [7, 27, 55]. A standard defense is to enforce Local Differential Privacy (LDP) [46]. In practice, clients first perform per-example ℓ_2 clipping before prototype computation to bound sensitivity, then add calibrated isotropic Gaussian noise when transmitting prototypes, a recipe we call Isotropic Gaussian Prototype Perturbation (IGPP). However, as illustrated in Figure 1, IGPP suffers from two core drawbacks. ❶ **Noise and discriminability mismatch.** Feature dimensions contribute unevenly to classification, with some carrying critical discriminability while others are largely redundant. Isotropic noise blindly over-perturbs the most informative dimensions and degrades class separability. ❷ ℓ_2 **clipping-threshold dilemma.** Prototype scales vary based on class size and domain characteristics. A larger threshold reduces clipping distortion but demands massive noise injection, whereas a smaller threshold forces excessive shrinkage, irrevocably erasing semantic content.

Driven by these challenges, we introduce two modules that jointly account for prototype geometry and privacy. **First**, to address ❶, we derive dimension-wise discriminative scores from the intra- and inter-class variability of sample embeddings and use them to privately select a discriminative subspace. We then perform groupwise clipping on each embedding, construct prototypes from the clipped representations, and inject group-specific noise to reallocate the privacy budget toward task-relevant coordinates for better representation quality. **Second**, to tackle ❷, we append a differentiable soft-clipping layer to the local encoder and enforce prediction consistency between pre- and post-clipped features via knowledge distillation. This guidance breaks the norm–weight compensation shortcut and drives feature norms to concentrate near the clipping threshold, reducing information loss under the same privacy budget.

In this paper, we propose a client-side privacy plug-in named **VPDR**, which comprises **V**ariance-adaptive **P**rototype Perturbation (VPP) and **D**istillation-guided Clipping **R**egularization (DCR). We leverage VPP to steer massive noise away from discriminative dimensions, preserving task-relevant structure while avoiding over-perturbation. We also employ DCR, designing a consistency regularizer that mitigates clipping-induced artifacts while keeping the baseline task loss unchanged. This synergy enables ProtoPFL frameworks equipped with VPDR to achieve more robust personalization than IGPP-based baselines. Formal analysis and systematic evaluations under two privacy attacks further show that VPDR provides LDP guarantees no weaker than those of IGPP while achieving a superior privacy–utility trade-off. Our main contributions are:

- We fuse intra- and inter-class variance to quantify per-dimension discriminability, yielding, to our knowledge, the first dimension-wise adaptive protection mechanism for prototype release with LDP guarantees.
- We design a distillation-guided regularization that couples soft clipping with distillation consistency, suppressing clipping artifacts while enhancing both robustness and representation fidelity.
- Integrated into existing ProtoPFLs, VPDR yields consistent gains on three multi-domain benchmarks, with ablations and privacy attacks supporting the effectiveness of both VPP and DCR. Code is available at `https://github.com/yuCoryx/ProtoPFL_VPDR`.

2. Related Work

Differential Privacy in FL. A large body of work studies DP for gradients or model updates in standard FL [3, 5, 12, 32, 48]. DP-FedAvg [12] applies server-side Gaussian noise to achieve client-level central DP, while BLUS+LUS [5] and DP-FedSAM [32] reduce the accuracy degradation via gradient sparsification and flatness-aware training. Extensions to PFL include PPSGD [3] and FedDPA [48] adaptively mix or regularize personalized parameters under joint DP guarantees. To guard against untrusted servers, LDP mechanisms have also been introduced [6, 14, 37, 41, 46]. UDP-FL [46] perturbs each user's update with calibrated Gaussian noise, and later schemes such as ACS-FL [14], FedFR-ADP [41], and ALDP-FL [6] refine this idea with adaptive clipping, noise scaling, and budget control, while all providing LDP guarantees for each client.

Prototype-based PFL. ProtoPFL uses class prototypes to align feature spaces across clients. FedProto [36] mitigates heterogeneity by exchanging prototypes, and FedPCL [11] improves semantic consistency through prototype-level contrastive learning. FPL [15], FedGMKD [54], and FedPLVM [42] integrate clustering and sparsity regularization to enhance robustness and generalization. More recent methods, including FedTGP [53], FedKTL [52], and MPFT [55], leverage server-side training and knowledge transfer to reinforce global semantics. Together, these advances shift ProtoPFL from static prototype sharing toward dynamic semantic alignment, improving personalization across clients. Despite growing interest in privacy for prototype release, existing defenses [11, 42, 55] often inject isotropic Gaussian noise into prototypes—typically without explicit per-example clipping or a full local DP accounting. As a result, *they lack formal DP guarantees and overlook task-relevant prototype geometry, thereby degrading representation quality*. We present the first LDP mechanism for ProtoPFL that combines groupwise clipping with adaptive noise allocation, offering privacy guarantees no weaker than isotropic perturbation and a better privacy–utility trade-off.

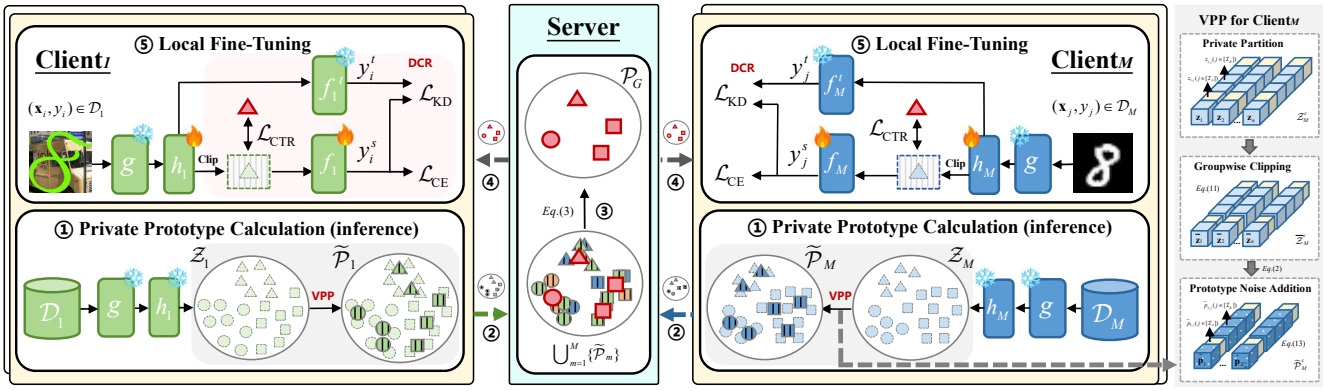


Figure 2. **Architecture illustration** of the ProtoPFL with VPDR. ① Private Prototype Calculation: each client runs VPP (Section 4.2) to privately partition embeddings, apply groupwise clipping, and add adaptive noise to prototypes; ② Upload: clients send privatized prototypes; ③ Global Prototype Generation: the server aggregates or trains global prototypes; ④ Download: clients receive global prototypes; ⑤ Local Fine-Tuning: each client fine-tunes locally with DCR (Section 4.3) for feature-norm stabilization alongside the task loss.

3. Preliminaries

3.1. Local Differential Privacy

We adopt client-side Local Differential Privacy (LDP) on each client's dataset [37, 44, 46] in this work, which protects individual training examples from an untrusted server.

Definition 3.1 ((ϵ, δ)–LDP [37, 46])**.** *A randomized mechanism $\mathcal{M} : \mathcal{X} \to \mathcal{R}$ satisfies (ϵ, δ)–LDP if for any adjacent $\mathcal{D}$ and $\mathcal{D}'$ differing by one sample and measurable $\mathcal{S} \subseteq \mathcal{R}$, $\Pr[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \leq e^{\epsilon} \Pr[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta$, where ϵ bounds the privacy loss and $\delta \in [0, 1)$ allows a failure probability.*

Definition 3.2 (Sensitivity [46])**.** *Given a function f, the ℓ_p sensitivity of f is defined as $\Delta_f = \max_{\mathcal{D} \sim \mathcal{D}'} \|f(\mathcal{D}) - f(\mathcal{D}')\|_p$, where $\|\cdot\|_p$ is the ℓ_1 or ℓ_2 norm.*

Definition 3.3 (Gaussian mechanism [46])**.** *For ℓ_2 sensitivity Δ_f, the Gaussian mechanism releases $\mathcal{M}(\mathcal{D}) = f(\mathcal{D}) + \mathcal{N}\big(0, (\sigma\Delta_f)^2 \mathbf{I}_d\big)$, which satisfies (ϵ, δ)–LDP for a suitable noise multiplier $\sigma > 0$.*

Theorem 3.1 (Noise calibration [46])**.** *In T-round FL, suppose each client m releases $\widetilde{s}_m = s_m + \mathcal{N}\big(0, (\sigma\Delta_s)^2 \mathbf{I}_d\big)$, where $s_m = f(\mathcal{D}_m)$ has ℓ_2 sensitivity Δ_s. Then there exists a constant $c_1 > 0$ such that $\sigma \geq c_1 \sqrt{T \ln(1/\delta)}/\epsilon$ ensures (ϵ, δ)–LDP over T rounds for each client m.*

3.2. Oneshot Laplace Mechanism

The oneshot Laplace mechanism selects Top-k coordinates by adding Laplace noise once and ranking the perturbed scores, ensuring DP without iterative peeling [28].

Definition 3.4 (Oneshot Laplace for Top-k [28])**.** *Given queries $f_1, \ldots, f_d$ on the dataset $\mathcal{D}$ with common ℓ_1-sensitivity Δ_f, draw $b_i \sim \mathrm{Lap}(\lambda)$ and set $y_i = f_i(\mathcal{D}) + b_i$. The mechanism then outputs the indices of the k largest y_i's.*

Theorem 3.2 (Oneshot Laplace Privacy [28])**.** *If $\lambda \geq 2k\,\Delta_f/\epsilon$, then the oneshot Laplace mechanism is $(\epsilon, 0)$–DP.*

4. Methodology

4.1. Problem Statement and Motivation

Personalized Federated Fine-Tuning. A federated system has M clients, each client $m \in [M]$ with dataset $\mathcal{D}_m$ of size n_m drawn from a domain-shifted distribution $P_m(\mathbf{x} \mid y)$. PFL seeks a client-specific model F_m that minimizes:

$$\min_{F_m} \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}_m} \mathcal{L}\big(F_m(\mathbf{x}), y\big), \tag{1}$$

where $\mathcal{L}(\cdot, \cdot)$ denotes the local loss. We write $F_m = f_m \circ h_m$, with a frozen backbone g and a trainable adapter A_m in feature extractor h_m, followed by a classifier f_m.

Prototype-based Personalization. Each client maps private samples to d-dimensional embeddings via h_m, yielding class-wise sets $\mathcal{Z}_m^c = \{\mathbf{z}_i^c = h_m(\mathbf{x}_i^c) \mid \mathbf{x}_i^c \in \mathcal{D}_m^c\}$. For class c, client m calculates a class-wise prototype set:

$$\{\mathbf{p}_m^{c,k}\} \leftarrow \mathrm{Calculate}\big(\{\mathbf{z}_i^c\}\big),\ \mathcal{P}_m^c = \big\{\mathbf{p}_m^{c,k}\big\}_{k=1}^{K_m^c}, \tag{2}$$

where K_m^c is the number of local prototypes for class c on client m. $K_m^c = 1$ recovers a single class mean [11, 15, 36, 53], while $K_m^c > 1$ yields multiple local centroids [42, 54, 55]. The server aggregates $\{\mathcal{P}_m^c\}_{m=1}^M$ and generates:

$$\{\mathbf{p}_g^{c,k}\} \leftarrow \mathrm{Generate}\Big(\cup_{m=1}^M \mathcal{P}_m^c\Big),\ \mathcal{P}_g^c = \big\{\mathbf{p}_g^{c,k}\big\}_{k=1}^{K_g^c}, \tag{3}$$

where K_g^c is the number of global prototypes per class. $K_g^c = 1$ denotes a global prototype obtained either by averaging [11, 36] or by training from client prototypes [53], and $K_g^c > 1$ corresponds to clustering into multiple global centroids [15, 42]. The global set $\mathcal{P}_g = \{\mathcal{P}_g^c\}$ is then broadcast and used as global anchors in existing contrastive or classification objectives for multi-domain alignment.

Isotropic Gaussian Prototype Perturbation (IGPP). To achieve client-side LDP, each client (i) applies per-example ℓ_2 clipping before any averaging or clustering,

$$\overline{\mathbf{z}}_i^c = \mathbf{z}_i^c \cdot \min\Big(1,\ R/\|\mathbf{z}_i^c\|_2\Big), \tag{4}$$

(ii) compute prototypes from $\{\overline{\mathbf{z}}_i^c\}$, and (iii) privatize each prototype release by adding isotropic Gaussian noise:

$$\widetilde{\mathbf{p}} = \mathbf{p} + \boldsymbol{\xi},\ \boldsymbol{\xi} \sim \mathcal{N}\big(\mathbf{0}, (\sigma_{\text{iso}}\Delta)^2 \mathbf{I}_d\big) \tag{5}$$

where Δ equals $2R/n_m^c$ for class means and $2R/n_m^{c,k}$ for clustering of size $n_m^{c,k}$ within class c. The noise multiplier is calibrated to (ϵ, δ)–LDP over T rounds via

$$\sigma_{\text{iso}} \geq c_1 \sqrt{T \ln(1/\delta)}/\epsilon \quad \text{(Theorem 3.1)}. \tag{6}$$

However, IGPP leaves two practical issues. **First**, applying isotropic noise perturbs all coordinates equally and can over-distort the most discriminative directions. **Second**, choosing a fixed ℓ_2 clipping threshold is delicate. A large threshold inflates noise via Δ while a small one severely distorts feature semantics. To address both, we propose **VPDR**, a client-side plug-in for ProtoPFL. As sketched in Figure 2, our **V**ariance-adaptive **P**rototype **P**erturbation (**VPP**) redistributes noise across dimensions to minimize information loss while preserving the same LDP guarantee, and our **D**istillation-guided **C**lipping **R**egularization (**DCR**) on each local adaptation stabilizes the per-sample norm bound. A summary of notation is provided in Appendix B.

4.2. Variance-Adaptive Prototype Perturbation

Isotropic noise overlooks the fact that feature dimensions matter differently for recognition: perturbing more informative dimensions harms utility more than redundant ones. Intuitively, dimensions with low intra-class variance often encode stable class-specific semantics, whereas those with high inter-class variance are more effective at separating classes. This motivates identifying an informative subspace and assigning it less perturbation. The main challenge, however, is that selecting such a subspace is itself data-dependent and therefore privacy-sensitive. We then split the budget over T communication rounds via sequential composition: $(\epsilon, \delta) = (\epsilon_1, 0) + (\epsilon_2, \delta)$. Specifically, we allocate $\epsilon_1 = r\epsilon$ with $r \in (0, 1)$ to subspace selection and reserve $\epsilon_2 = (1 - r)\epsilon$ for prototype release (allocation diagram in Appendix C). This yields VPP, which adaptively reallocates perturbation while preserving the overall privacy guarantee.

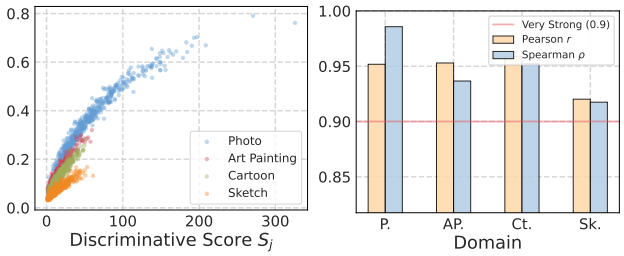
Discriminative Partition. For each $j \in [d]$, we define

$$V_j^{\text{tra}} = \sum_c (n_m^c - 1) s_{c,j}^2,\ V_j^{\text{ter}} = \sum_c n_m^c (\mu_{c,j} - \mu_j)^2, \tag{7}$$

where the intra-class variance $s_{c,j}^2$, the class mean $\mu_{c,j}$ and the client-level mean μ_j are computed from encodings $\mathcal{Z}_m$. The ANOVA-normalized [30] discriminative score is:

$$S_j = \frac{V_j^{\text{ter}}/(C-1)}{V_j^{\text{tra}}/(n_m - C) + \zeta},\ \zeta > 0. \tag{8}$$

This normalization mitigates the effects of class-size imbalance and small-sample bias. *Empirically*, we find that S_j exhibits a strong linear correlation with the label mutual information $I(z_j; y)$ (Figure 3a), with both Pearson and Spearman coefficients consistently exceeding 0.90 across domains (Figure 3b). This makes S_j a reliable and lightweight proxy for dimension-wise discriminability.



(a) S_j vs. $I(z_j; y)$ scatter. (b) Per-domain correlations.

Figure 3. **Correlation** between the discriminative score S_j and label mutual information $I(z_j; y)$ on PACS.

Although never uploaded, the selected index set determines the groupwise clipping bounds and noise covariance. Thus, *the released prototypes remain a function of this private selection* and must be included in privacy accounting. To privately isolate informative dimensions, we clip raw scores to bound the ℓ_1 sensitivity, $\overline{S}_j = \text{clip}(S_j, 0, H) := \min\{\max(S_j, 0),\ H\}$, and apply the oneshot Laplace Top-k mechanism [28]. Setting $k = d_A = \lceil \rho d \rceil$ for $\rho \in (0, 0.5]$ isolates the discriminative subspace $\mathcal{I}_A$, leaving the rest $d_B = d - d_A$ as the non-discriminative subspace $\mathcal{I}_B$. Calibrating the per-round Laplace noise scale as

$$\lambda \geq 2 d_A H T / \epsilon_1 \tag{9}$$

guarantees $(\epsilon_1, 0)$-LDP for the partition step over T rounds as established by Theorem 4.1.

Privatized Prototype Release. Given this partition, we decompose the feature vector as $\mathbf{z} = [\mathbf{z}_A; \mathbf{z}_B]$, where $\mathbf{z}_A = \mathbf{z}[\mathcal{I}_A]$ and $\mathbf{z}_B = \mathbf{z}[\mathcal{I}_B]$. For a fixed global clipping threshold R, we allocate groupwise bounds:

$$R_A = R\kappa_A, R_B = R\kappa_B, \tag{10}$$

where $\kappa_A = \sqrt{d_A/d}, \kappa_B = \sqrt{d_B/d}$, so that $R_A^2 + R_B^2 = R^2$. We then independently apply groupwise ℓ_2 clipping:

$$\begin{aligned} \overline{\mathbf{z}}_A &= \mathbf{z}_A \cdot \min\big(1, R_A/\|\mathbf{z}_A\|_2\big), \\ \overline{\mathbf{z}}_B &= \mathbf{z}_B \cdot \min\big(1, R_B/\|\mathbf{z}_B\|_2\big), \end{aligned} \tag{11}$$

and form prototypes $\mathbf{p} = [\mathbf{p}_A; \mathbf{p}_B]$ by computing the mean or clustering on each group. This yields group sensitivities $\Delta_A = 2R_A/n_m^c = \Delta\kappa_A$ and $\Delta_B = 2R_B/n_m^c = \Delta\kappa_B$.

Let σ_{ref} denote the noise multiplier of the reference isotropic Gaussian mechanism calibrated to (ϵ_2, δ). Theorem 4.2 establishes that groupwise reallocation can be performed without weakening the privacy guarantee, provided that $1/\sigma_A^2 + 1/\sigma_B^2 \leq 1/\sigma_{\text{ref}}^2$. We satisfy it by introducing weights $w_A, w_B > 0$ with $w_A + w_B = 1$ and parameterizing the groupwise multipliers as

$$\sigma_A = \sigma_{\text{ref}}/\sqrt{w_A},\ \sigma_B = \sigma_{\text{ref}}/\sqrt{w_B}. \tag{12}$$

For a parameter-free design, we set these weights solely

based on the group dimensions:

$$w_A = \kappa_B/(\kappa_A + \kappa_B), w_B = 1 - w_A, \tag{13}$$

The privatized prototype release is then formulated as:

$$\widetilde{\mathbf{p}} = \big[\mathbf{p}_A + \boldsymbol{\xi}_A,\ \mathbf{p}_B + \boldsymbol{\xi}_B\big], \tag{14}$$

$$\boldsymbol{\xi}_A \sim \mathcal{N}\big(\mathbf{0}, (\sigma_A \Delta_A)^2 \mathbf{I}_{d_A}\big), \boldsymbol{\xi}_B \sim \mathcal{N}\big(\mathbf{0}, (\sigma_B \Delta_B)^2 \mathbf{I}_{d_B}\big).$$

Under this construction, the release preserves the same (ϵ_2, δ)-LDP guarantee as the reference mechanism.

Utility–privacy trade-off. Under the sequential composition $(\epsilon, \delta) = (\epsilon_1, 0) + (\epsilon_2, \delta)$, reserving budget for subspace selection implies $\epsilon_2 < \epsilon$, which increases the reference multiplier σ_{ref}. VPP offsets this by actively steering noise away from task-relevant coordinates. Specifically, ensuring the informative subspace $\mathcal{I}_A$ is *no noisier* than the isotropic reference requires $\sigma_A \Delta_A \leq \sigma_{\text{ref}} \Delta$. Substituting our weights (13) and sensitivities, we obtain:

$$\kappa_A \sqrt{(\kappa_A + \kappa_B)/\kappa_B} \leq 1 \ \Leftrightarrow\ \kappa_A \leq \kappa_B \ \Leftrightarrow\ 0 < \rho \leq 0.5.$$

Consequently, restricting the selected discriminative dimensions to at most half of the feature space ensures that $\mathcal{I}_A$ receives no more noise than the isotropic (ϵ_2, δ)-LDP mechanism, while the excess noise is shifted to the redundant subspace $\mathcal{I}_B$. As shown in Section 5, this targeted preservation of discriminative features delivers a net utility gain that outweighs the overhead induced by privacy budget splitting.

4.3. Distillation-Guided Clipping Regularization

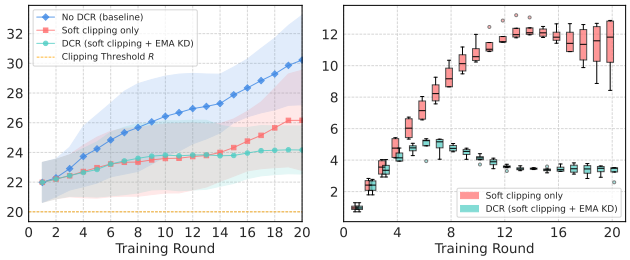
To ensure bounded sensitivity prior to adding calibrated Gaussian noise, clients typically apply per-example ℓ_2 feature clipping before prototype construction. Hard clipping, however, is brittle: when a feature norm significantly exceeds the threshold ($\|\mathbf{z}_i\|_2 \gg R$), hard clipping aggressively shrinks the vector and irrevocably erases structural information. Conversely, for undersized features ($\|\mathbf{z}_i\|_2 \ll R$), the clipping operation is entirely ineffective, leaving small-magnitude vectors vulnerable to the injected noise. DCR addresses this dilemma by regularizing the feature space dynamically during local training.

Soft Clipping. To mitigate sensitivity to R in the prototype perturbation phase, we append a differentiable soft-clipping layer at the end of the feature encoder during local training:

$$\widehat{\mathbf{z}}_i = \frac{R}{\|\mathbf{z}_i\|_2 + \gamma R} \cdot \mathbf{z}_i, \quad 0 < \gamma \ll 1, \tag{15}$$

where γ controls the strength. This expands small-norm features toward R and smoothly contracts large-norm features, avoiding the gradient discontinuities of hard clipping.

Consistency Guidance. A potential failure mode is that the model counteracts soft clipping by inflating the classifier weights. As Figure 4a shows, pre-clipping feature norms tend to grow during mid-to-late training under soft clipping only. To decouple this shortcut, we introduce an Exponential Moving Average (EMA) teacher–student distillation that enforces prediction consistency between pre- and post-clipped features without parameter coupling. Concretely, we split the local classifier into a trainable student head f_m and a momentum teacher f_m^t updated as



(a) Pre-clipping feature norm. (b) Teacher–student logit difference.

Figure 4. **Evaluation of feature norm and logit difference** of FedPLVM [42] with IGPP on Office–Caltech.

$$\theta_m^t = \beta \theta_m^t + (1 - \beta)\theta_m, \tag{16}$$

where θ and θ_m^t denote the student and teacher parameters, respectively, and $\beta \in [0, 1)$ is the EMA momentum. At each training step the teacher produces soft targets on the original feature, $y^t = f_m^t(\mathbf{z})$, while the student predicts on the soft-clipped feature, $y^s = f_m(\overline{\mathbf{z}})$. We minimize the KL divergence between their temperature-scaled distributions:

$$\mathcal{L}_{\text{KD}} = \text{KL}\Big(\text{softmax}\big(\frac{y^t}{\tau}\big) \big\| \text{softmax}\big(\frac{y^s}{\tau}\big)\Big), \tag{17}$$

where τ is the distillation temperature. This consistency prevents the model from bypassing clipping via weight rescaling, effectively breaking the norm–weight tug-of-war.

Let $\mathcal{L}_{\text{BASE}}$ denote the ProtoPFL objective (e.g., cross-entropy for classification $\mathcal{L}_{\text{CE}}$ or an InfoNCE-style contrastive alignment $\mathcal{L}_{\text{CTR}}$ [11, 15, 42, 53]) as described in Figure 2. The overall local fine-tuning objective is

$$\mathcal{L} = \mathcal{L}_{\text{BASE}} + \lambda_1 \mathcal{L}_{\text{KD}}. \tag{18}$$

Here λ_1 is the KD weight. *Empirically*, DCR consistently reduces norm drift and concentrates pre-clipping feature norms near R during the mid-to-late training (Figure 4a). Also, it lowers the teacher–student logit discrepancy, providing evidence of stronger prediction consistency (Figure 4b). These behaviors make the framework less sensitive to the exact choice of R while preserving task semantics.

4.4. ProtoPFL with VPDR

Algorithm 1 outlines ProtoPFL with VPDR. In each round, clients first privately select a discriminative subspace and release privatized prototypes for uplink transmission. The server aggregates or trains these representations into global prototypes and broadcasts them back. With the global prototypes fixed, clients conduct local personalized fine-tuning via DCR while optimizing the base ProtoPFL objective.

Privacy Guarantees. VPDR confines all privacy-sensitive operations to the VPP during uplink transmission. Conversely, DCR is a purely local regularization that con-

Algorithm 1 ProtoPFL with VPDR

Input: Communication rounds T , local epochs E, classes C , clients M, private dataset $\mathcal{D}_m = \{\mathbf{x}_i, y_i\}_i^{n_m}$, global clipping threshold R, global noise multiplier σ
Output: Well-trained client models

for $t = 1, 2, \cdots, T$ **do**
  // Local Private Prototype Calculation
  for Client $m = 1, 2, \ldots, M$ **do**
    Encode features $z_i^c = h_m(\mathbf{x}_i^c)$
    Compute dimension scores S via (7) and (8);
    Privately partition to obtain $\mathcal{I}_A$ and $\mathcal{I}_B$;
    Perform groupwise clipping $\{\overline{z}_i^c\}$ by (11);
    for Class $c = 1, 2, \ldots, C$ **do**
      Get local prototypes and add noise by (14).
    end for
    Upload the privatized prototype set $\widetilde{\mathcal{P}}_m$.
  end for
  // Global Prototype Generation
  Server aggregates or trains the global set $\mathcal{P}_g$ by (3).
  // Local Personalized Fine-Tuning
  for Client $m = 1, 2, \ldots, M$ **do**
    for $e = 1, 2, \cdots, E$ **do**
      Apply soft clipping to features via (15);
      Update the EMA teacher via (16);
      Update adapter and classifier by Eq. (18).
    end for
  end for
end for

sumes no privacy budget. Under the budget split in Section 4.2, VPP is analyzed in two steps: Theorem 4.1 provides $(\epsilon_1, 0)$–LDP for private subspace selection, and Theorem 4.2 provides (ϵ_2, δ)–LDP for prototype release. By sequential composition, ProtoPFL with VPDR satisfies overall (ϵ, δ)–LDP. Full proofs are deferred to Appendix D.

Theorem 4.1 (Partitioning privacy)**.** *Let $\{S_j\}_{j=1}^d$ be the scores from (8) computed on client m, and define $\overline{S}_j = \mathrm{clip}(S_j, 0, H)$ under adjacency on $\mathcal{D}_m$. In each round, execute the one-shot Laplace Top-k mechanism ($k = d_A$) by drawing $b_j \sim \mathrm{Lap}(\lambda)$ to form $y_j = \overline{S}_j + b_j$, outputting the indices of the top-k values. If $\lambda \geq 2d_A HT/\epsilon_1$, the T-round partitioning process satisfies $(\epsilon_1, 0)$-LDP for client m.*

Next, conditioned on the selected subspace, we calibrate noise for releasing the groupwise clipped prototypes.

Theorem 4.2 (Release privacy)**.** *Let the isotropic Gaussian release with sensitivity Δ achieve (ϵ_2, δ)–LDP after T rounds using multiplier $\sigma_{\mathrm{ref}} \geq c_2\sqrt{T\ln(1/\delta)}/\epsilon_2$. Under VPP, use groupwise covariance $\Sigma_{\mathrm{VPP}} = \mathrm{diag}((\sigma_A\Delta_A)^2\mathbf{I}_{d_A}, (\sigma_B\Delta_B)^2\mathbf{I}_{d_B})$ with $\Delta^2 = \Delta_A^2 + \Delta_B^2$. If the group multipliers satisfy*

$$\frac{1}{\sigma_A^2} + \frac{1}{\sigma_B^2} \leq \frac{1}{\sigma_{\mathrm{ref}}^2}, \tag{19}$$

then the T-round prototype release in VPP is also (ϵ_2, δ)–LDP for client m.

Remark 4.1. *A convenient choice satisfying (19) is $\sigma_A = \sigma_{\mathrm{ref}}/\sqrt{w}$ and $\sigma_B = \sigma_{\mathrm{ref}}/\sqrt{1-w}$ with $w \in (0, 1)$. In VPP, we set w according to (13), so the noise allocation is fully determined by the discriminability-aware partition and automatically adapts to the relative dimensionalities d_A and d_B, without introducing any extra tunable hyperparameter.*

Complexity and Overheads. VPDR introduces only modest overhead to existing ProtoPFLs. (1) **Computation:** Per communication round, VPP incurs an extra $\mathcal{O}(n_m d + d\log d)$ time over baseline construction, while noise addition remains $\mathcal{O}(d)$. DCR adds $\mathcal{O}(n_m dCE/B)$ training time for the teacher forward pass and KL divergence computation. (2) **Memory:** Overhead is minimal, requiring only $\mathcal{O}(d)$ to store the score vector and mask, alongside $\mathcal{O}(dC)$ for the momentum teacher head. (3) **Communication:** Complexity remains strictly unchanged at $\mathcal{O}(Cd)$, as clients solely transmit the final privatized prototypes.

5. Experiments

5.1. Experimental Setup

Datasets. We conduct experiments on three domain-skew benchmarks: Digits [16, 18, 26, 29], which comprises MNIST (MNI.), USPS, SVHN and synthetic digits (SYN); Office–Caltech [13], consisting of Caltech (Cal.), Amazon (Amz.), DSLR (DSR.) and Webcam (Web.); PACS [20] covering Photo (P.), Art Painting (AP.), Cartoon (Ct.) and Sketch (Sk.). Details are provided in Appendix E.1.

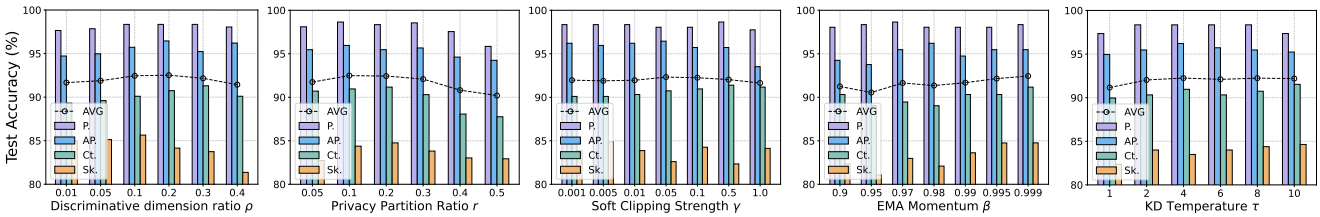
Model. All experiments use a ViT-small [8] backbone with a hidden size d of 512 and frozen pretrained weights. We insert lightweight adapters after each Transformer block, and train only the adapters and the classifier.

Counterparts. We compare six ProtoPFL frameworks, all equipped with IGPP: FedProto [36], FedPCL [11], FPL [15], FedPLVM [42], FedTGP [53], and MPFT [55]. Personalization is achieved by exchanging only privatized prototypes. For MPFT, the server learns an adapter from the privatized prototypes and transmits it to clients, which is the post-processing of DP [9] and no extra privacy cost.

Implementation Details. Each benchmark has four domain-specific clients [11, 42], training on 10% of Digits and 30% of Office–Caltech and PACS. We run communication rounds $T = 20$ with local epochs $E = 2$ and batch size 256. Local optimization is AdamW with learning rate 10^{-3}, weight decay 10^{-5}, and momentum 0.9. ProtoPFL hyperparameters are listed in Appendix E.2. Unless otherwise noted, we set (ϵ, δ)=$(1, 10^{-5})$ and select the clipping radius R by grid search over $\{5, 10, 15, 20\}$. VPDR defaults are $H = 0.1$, $r = 0.1$, $\rho = 0.2$, $\gamma = 0.05$, $\beta = 0.999$,

Table 1. **Comparison** of Average Accuracy (AVG) and Standard Deviation (STD) under domain skew. The **best** is marked.

Methods		Digits						Office–Caltech						PACS					
Framework	+LDP	MNI.	USPS	SVHN	SYN	AVG↑	STD↓	Amz.	Cal.	DSR.	Web.	AVG↑	STD↓	P.	AP.	Ct.	Sk.	AVG↑	STD↓
FedProto [36]	+IGPP	98.20	93.42	89.80	96.00	94.36	3.45	95.32	91.07	81.38	95.51	90.82	6.62	97.70	92.71	91.52	80.37	90.58	7.31
	+VPDR	98.31	94.87	92.33	98.70	**96.05**	**2.78**	96.71	93.55	84.85	98.32	**93.36**	**6.01**	98.80	95.60	91.88	84.59	**92.71**	**6.11**
FedPCL [11]	+IGPP	98.30	90.08	91.54	95.35	93.82	3.41	95.29	92.41	64.52	94.92	86.79	13.53	97.31	85.33	80.34	55.92	79.73	16.08
	+VPDR	98.92	92.63	93.15	96.95	**95.41**	**2.63**	96.33	92.86	74.97	96.61	**90.19**	**8.90**	98.50	91.93	87.18	68.03	**86.91**	**11.84**
FPL [15]	+IGPP	98.15	93.67	91.89	96.45	95.04	2.44	94.76	92.36	82.15	96.31	91.40	6.37	98.10	94.13	91.16	82.08	91.37	6.28
	+VPDR	98.10	95.60	93.40	97.60	**96.20**	**2.35**	96.34	93.72	85.02	99.65	**93.68**	**6.36**	98.86	96.78	93.15	83.93	**93.18**	**5.86**
FedPLVM [42]	+IGPP	97.81	90.43	92.57	92.20	93.25	2.88	95.61	91.26	82.05	96.61	91.38	6.64	97.90	92.15	84.11	74.20	87.09	9.34
	+VPDR	97.71	92.93	94.57	93.70	**94.73**	**1.79**	97.30	92.23	84.76	99.21	**93.38**	**6.07**	98.80	93.71	86.04	77.63	**89.05**	**8.38**
FedTGP [53]	+IGPP	97.62	93.87	91.69	97.20	95.10	2.44	96.34	92.38	83.97	96.55	92.31	**5.88**	98.80	96.62	92.09	84.29	92.95	5.80
	+VPDR	97.84	95.32	93.24	98.75	**96.29**	**2.14**	97.34	96.61	86.21	100	**95.04**	6.19	100	97.32	94.83	87.01	**94.79**	**5.64**
MPFT [55]	+IGPP	97.61	94.27	92.95	98.00	95.71	**2.16**	95.76	92.52	84.03	96.61	92.23	**5.74**	99.63	96.33	93.16	84.50	93.41	5.89
	+VPDR	98.64	94.77	93.78	99.05	**96.56**	2.32	97.81	94.15	86.58	98.31	**94.71**	5.83	99.40	97.70	95.65	86.88	**94.91**	**5.04**



(a) The effect of ρ. (b) The effect of r. (c) The effect of γ. (d) The effect of β. (e) The effect of τ.

Figure 5. **Hyperparameter sensitivity** of FedProto [36] framework with VPDR on PACS.

Table 2. **Ablation results** of FedProto [36] framework.

VPP	DCR	Office-Caltech					
		Amz.	Cal.	DSR	Web.	AVG↑	STD ↓
		95.32	91.07	81.38	95.51	90.82	6.62
✓		96.96	92.88	84.53	97.85	93.06	6.08
	✓	97.08	92.60	84.57	97.31	92.89	**5.96**
✓	✓	96.71	93.55	84.85	98.32	**93.36**	6.01
VPP	DCR	PACS					
		P.	AP.	Ct.	Sk.	AVG↑	STD ↓
		97.70	92.71	91.52	80.37	90.58	7.31
✓		98.91	93.15	92.18	83.31	91.89	6.44
	✓	98.62	93.70	91.82	83.51	91.91	6.29
✓	✓	98.80	95.60	91.88	84.59	**92.71**	**6.11**

Table 3. **Average accuracy** (%) on CIFAR-10 under label skew.

Methods		$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 1$	$\alpha = 5$	$\alpha = 10$
FedProto [36]	+IGPP	42.76	82.31	**90.23**	96.15	96.30
	+VPDR	**44.14**	**85.00**	89.99	**96.34**	**96.55**
FedPCL [11]	+IGPP	18.22	73.63	87.78	95.55	**95.67**
	+VPDR	**20.92**	**73.98**	**88.29**	**95.63**	95.61
FPL [15]	+IGPP	28.42	69.14	88.19	95.39	95.58
	+VPDR	**37.18**	**80.14**	**88.73**	**96.04**	**96.29**
FedPLVM [42]	+IGPP	37.33	76.95	**87.21**	95.06	95.32
	+VPDR	**40.17**	**77.42**	82.76	**95.47**	**95.34**
FedTGP [53]	+IGPP	40.71	79.92	87.68	95.28	95.47
	+VPDR	**43.03**	**82.46**	**89.61**	**96.32**	**96.42**
MPFT [55]	+IGPP	46.79	79.06	91.23	**96.64**	**96.71**
	+VPDR	**47.33**	**80.17**	**91.26**	96.53	96.64

$\tau = 4$ and λ_1= 0.05. We evaluate each personalized model on its own local data and report the average test accuracy.

5.2. Performance Comparison

Table 1 shows that VPDR consistently improves average accuracy (AVG) compared with IGPP across all ProtoPFLs and benchmarks. Improvements are most pronounced on the more heterogeneous PACS and Office–Caltech. While a few per-domain scores fluctuate, the per-dataset averages for every framework increase, and dispersion (STD) typically shrinks on harder regimes, indicating that VPDR's variance-adaptive noise and clipping stabilization primarily reduce multi-domain fluctuation where it matters most. Additional results and t-SNE plots are in Appendix E.3.

5.3. Diagnostic Analysis

Hyperparameter Study. Figure 5 illustrates the impact of hyperparameters on PACS in FedProto [36]. The optimal range for the privacy partition ratio r lies between 0.1 and 0.2, while the discriminative subspace ratio ρ performs best around 0.2. The optimal soft clipping strength γ is 0.05, and the EMA momentum β achieves its best performance at 0.999. The KD temperature τ is most effective at 4, with little variation between 4 and 8. These trends confirm that VPDR delivers robust performance across a wide spectrum and that our selected defaults lie close to the optimum.

Ablation Study. We ablate VPP and DCR within FedProto [36] as shown in Table 2. Starting from IGPP, adding either VPP or DCR alone improves performance on both Office–Caltech and PACS. Using both together yields the best average accuracy, confirming complementary gains.

Scalability under Label Skew. We induce client-level label skew on CIFAR-10 [17] using a Dirichlet distribution with concentration $\alpha \in \{0.1, 0.5, 1, 5, 10\}$, where smaller α means higher skew. Table 3 ($M = 20$, $\epsilon = 1$) shows

Table 4. **Privacy Attack Results** of FedProto [36] on Office-Caltech under varying privacy budgets.

ϵ	LDP	Feature Reconstruction (FSH)			Membership Inference (MIA)			
		Cos Sim↓	cFFD↑	Top-1 Hit(%)↓	ROC-AUC→ 0.5	TPR@1%FPR↓	Advantage→ 0	F1 Score↓
-	NoLDP	0.9999±0.0000	257.7±65.0	100.00±0.00	0.6729±0.0124	0.0298±0.0200	0.4458±0.0170	0.7881±0.0082
1	+IGPP	0.6437±0.0022	2214.6±21.0	20.29±3.52	0.5079±0.0340	0.0043±0.0045	0.2669±0.0300	0.7317±0.0085
	+VPP	0.6456±0.0026	2197.6±18.2	21.22±4.04	0.4992±0.0293	0.0040±0.0050	0.2536±0.0279	0.7282±0.0119
	+VPDR	0.6456±0.0032	2207.1±27.7	20.62±3.33	0.5055±0.0275	0.0042±0.0044	0.2649±0.0299	0.7321±0.0109
2	+IGPP	0.6710±0.0025	2038.5±22.7	33.23±3.59	0.5081±0.0336	0.0066±0.0036	0.2855±0.0332	0.7399±0.0073
	+VPP	0.6706±0.0025	2040.7±26.6	34.44±5.83	0.5097±0.0228	0.0065±0.0016	0.2834±0.0244	0.7406±0.0084
	+VPDR	0.6718±0.0029	2029.4±26.0	33.75±2.90	0.5089±0.0299	0.0063±0.0031	0.2846±0.0309	0.7404±0.0117

that VPDR yields the largest gains under extreme skew ($\alpha \leq 0.5$). While this advantage narrows toward the IGPP as data becomes balanced, it highlights that VPP's discriminative focus and DCR's norm stabilization thrive under high heterogeneity. Furthermore, we verify VPDR's robustness under joint model–data heterogeneity and its generalization to text modality in Appendices E.4 and E.5.

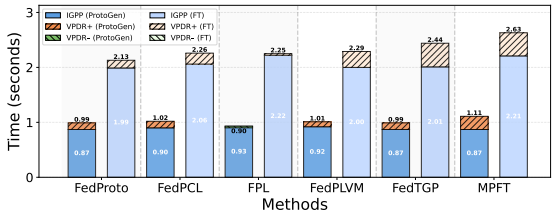
5.4. Defense Assessment Under Attacks

We consider a realistic deployment threat model in which an adversary can access client-uploaded prototypes, either as a semi-honest server or as an external eavesdropper. We evaluate two attacks against all clients per communication round: Feature-Space Hijacking (FSH) [55] and Membership Inference Attacks (MIA) [33]. FSH actively reconstructs raw client data by optimizing dummy inputs to match targeted prototypes. A successful defense yields low cosine similarity, high classifier feature fréchet distance (cFFD), and low Top-1 Hit rates. Conversely, MIA passively infers training set membership by thresholding sample-to-prototype distances, evaluated via ROC-AUC (where values near 0.5 indicate random guessing), TPR@1%FPR, advantage, and F1 score. Detailed attack settings are deferred to Appendix E.6.

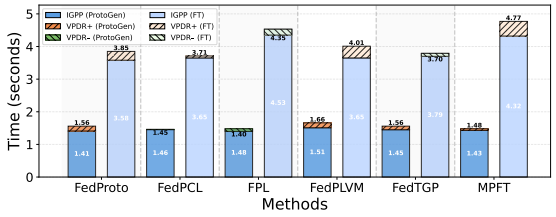
Table 4 reports the attack outcomes on Office-Caltech under $\epsilon \in \{1, 2\}$, with identical trends observed on Digits and PACS (Appendix E.6). Compared to NoLDP, all LDP mechanisms drastically degrade the attacker's capabilities. Crucially, VPP and VPDR defense metrics match the IGPP baseline within a single standard deviation. At $\epsilon = 1$, all three mechanisms drive MIA ROC-AUC to chance (~ 0.50) and throttle the FSH Top-1 Hit rate to roughly 20%. These results confirm that VPDR's variance-adaptive reallocation and soft clipping do not leak structural vulnerabilities, achieving superior utility under the same practical defense strength as isotropic Gaussian perturbation. More results varying the d and r are in Appendix E.7.

5.5. Computational Cost

Figure 6 reports the per-round, per-client wall-clock time on Office-Caltech and PACS, decomposed into Prototype Generation (ProtoGen) and Local Fine-Tuning (FT). Across six ProtoPFL frameworks, VPDR increases total runtime by an average of only 0.28s (8.1%) compared to the IGPP



(a) Office-Caltech.



(b) PACS.

Figure 6. **Per-client average time (s)** per communication round.

Table 5. Runtime (ms) of FedProto with VPDR on Office-Caltech.

Stage	Base	Extra VPDR	Overhead
ProtoGen	1057	13.22 (Stats) + 9.78 (Top-k) = 23.00	**2.2%**
FT	2055	1.58 (Clip) + 1.59 (Tch) + 1.38 (KL) = 4.55	**0.2%**

baseline, with phase-level increases of $+0.09$s for ProtoGen and $+0.20$s for FT. Table 5 details this cost for FedProto on Office-Caltech. VPDR operations incur marginal overheads: 2.2% in ProtoGen (variance statistics and Laplace Top-k) and 0.2% in FT (soft clipping, teacher-head forward, KL divergence term). Ultimately, VPDR delivers accuracy gains with acceptable or even negligible computational overhead and no additional server-side work.

6. Conclusion

We introduce VPDR, a client-side plug-in for privacy-preserving ProtoPFL that combines VPP, which reallocates noise across feature dimensions by discriminative strength, and DCR, which enforces soft-clipping and prediction consistency via knowledge distillation. Our analysis provides LDP guarantees no weaker than those of isotropic perturbation. Across diverse ProtoPFLs and multi-domain benchmarks, VPDR delivers higher utility at fixed privacy budgets and remains robust to label skew, while preserving communication cost and incurring only modest compute overhead. Evaluations against two attacks show near-chance performance, indicating strong protection.

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